Sorting Algorithms and Their Efficiency

Chapter 11

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Basic Sorting Algorithms

- Sorting:
  - Organize a collection of data into either ascending or descending order
- Internal sort
  - Collection of data fits in memory
- External sort
  - Collection of data does *not* all fit in memory
  - Must reside on secondary storage

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Basic Sorting Algorithms

- The Selection Sort
- The Bubble Sort
- The Insertion Sort

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The Selection Sort

- Select the largest item
  - Place in last position
- Select the next largest item
  - Place in 2nd to last position
- Repeat for remaining unsorted items
  - Requires several passes over the data

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The Selection Sort

Gray elements are selected;
blue elements comprise the sorted portion of the array.

Initial array:	29	10	14	37	13
After 1st swap:	29	10	14	13	**37**
After 2nd swap:	13	10	14	**29**	**37**
After 3rd swap:	13	10	**14**	**29**	**37**
After 4th swap:	**10**	**13**	**14**	**29**	**37**

- To sort an array into ascending order, you first search for the largest item.
- Place largest item in the last position of the array
- Swap the last item with the largest item, even if these items happen to be identical.
- Repeat

FIGURE 11-1 A selection sort of an array of five integers

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The Selection Sort

```
   for (int last = n - 1; last >= 1; last--)
   {
      // At this point, theArray[last+1..n-1] is sorted, and its
      // entries are greater than those in theArray[0..last].
      // Select the largest entry in theArray[0..last]
      int largest = findIndexofLargest(theArray, last+1);

      // Swap the largest entry, theArray[largest], with
      // theArray[last]
      std::swap(theArray[largest], theArray[last]);
   }  // end for
}  // end selectionSort

int findIndexofLargest(const ItemType theArray[], int size)
{
   int indexSoFar = 0;  // Index of largest entry found so far
   for (int currentIndex = 1; currentIndex < size; currentIndex++)
   {
      // At this point, theArray[indexSoFar] >= all entries in
      // theArray[0..currentIndex - 1]
      if (theArray[currentIndex] > theArray[indexSoFar])
         indexSoFar = currentIndex;
   }  // end for

   return indexSoFar;  // Index of largest entry
}  // end findIndexofLargest
```

Note this is a nested loop for n

LISTING 11-1 An implementation of the selection sort

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The Selection Sort

- Analysis
  - Selection sort is $O(n^2)$
  - Appropriate for small *n*
  - $O(n^2)$ grows rapidly
- Could be a good choice when
  - Data moves are costly,
  - But comparisons are not
- Pros
  - Easy to understand
  - Minimal additional space required
- Cons
  - $O(n^2)$ – slow
  - Does not preserve relative order of items with equal keys (not stable)
  - Potentially suitable for low volume only

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The Bubble Sort

- Compares adjacent items
  - Exchanges them if out of order
  - Requires several passes over the data
- When ordering successive pairs
  - Largest item bubbles to end of the array

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The Bubble Sort

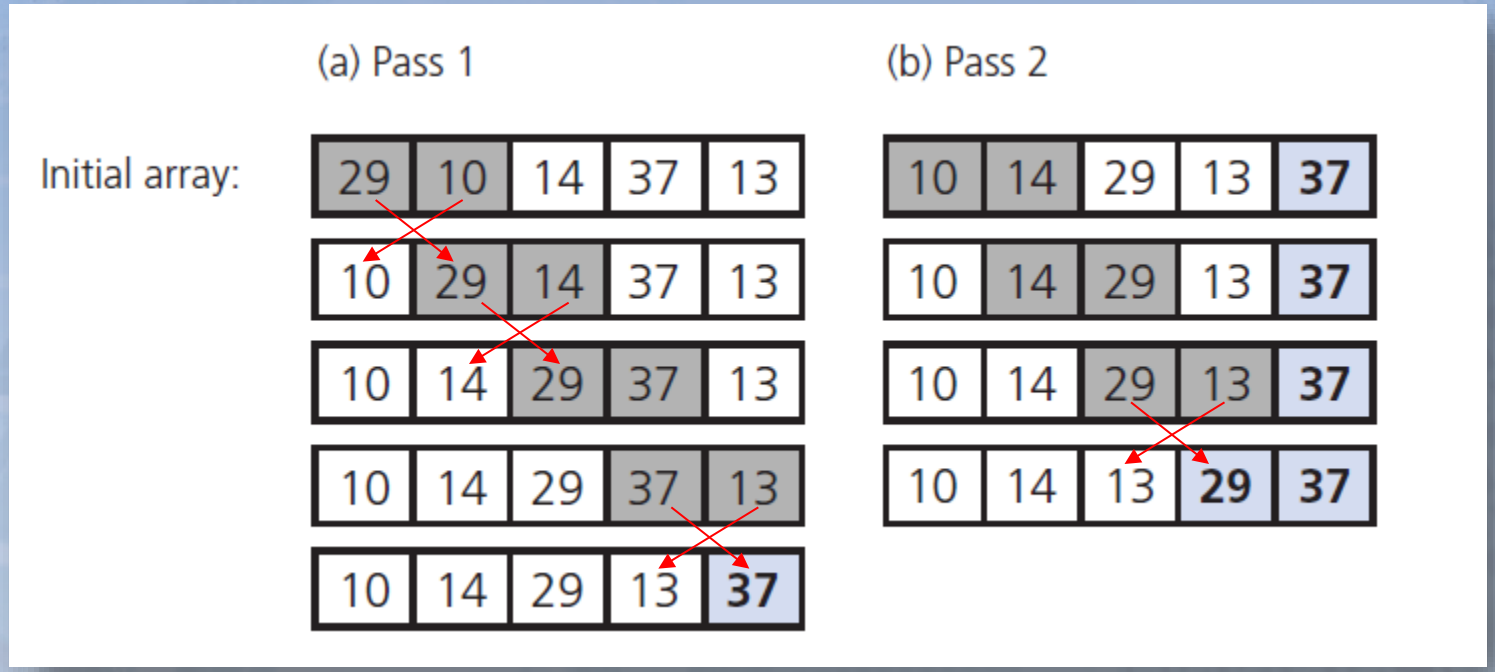

FIGURE 11-2 First two passes of a bubble sort of an array of five integers

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The Bubble Sort

LISTING 11-2 An implementation of the bubble sort

```
/** Sorts the items in an array into ascending order.
 @pre  None.
 @post  theArray is sorted into ascending order; n is unchanged.
 @param theArray  The given array.
 @param n  The size of theArray. */
void bubbleSort(ItemType theArray[], int n)
{
   bool sorted = false;     // False when swaps occur
   int pass = 1;
   while (!sorted && (pass < n))
   {
      // At this point, theArray[n+1-pass..n-1] is sorted
      // and all of its entries are > the entries in theArray[0..n-pass]
      sorted = true;        // Assume sorted
      for (int index = 0; index < n - pass; index++)
      {
         // At this point, all entries in theArray[0..index-1]
         // are <= theArray[index]
         int nextIndex = index + 1;
         if (theArray[index] > theArray[nextIndex])
         {
            // Exchange entries
            std::swap(theArray[index], theArray[nextIndex]);
            sorted = false;  // Signal exchange
         }  // end if
      }  // end for
      // Assertion: theArray[0..n-pass-1] < theArray[n-pass]

      pass++;
   }  // end while
}  // end bubbleSort
```

Note this is a nested loop for n

LISTING 11-2 An implementation of the bubble sort

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The Bubble Sort

- Analysis
  - Worst case $O(n^2)$
  - Best case (array already in order) is $O(n)$
- Pros
  - Easy to understand
  - No additional space required
  - Preserves relative order of items with equal keys (stable)
- Cons
  - $O(n^2)$ – slow
  - Potentially suitable for low volume only

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The Insertion Sort

- Take each item from unsorted region
  - Insert it into correct order in sorted region

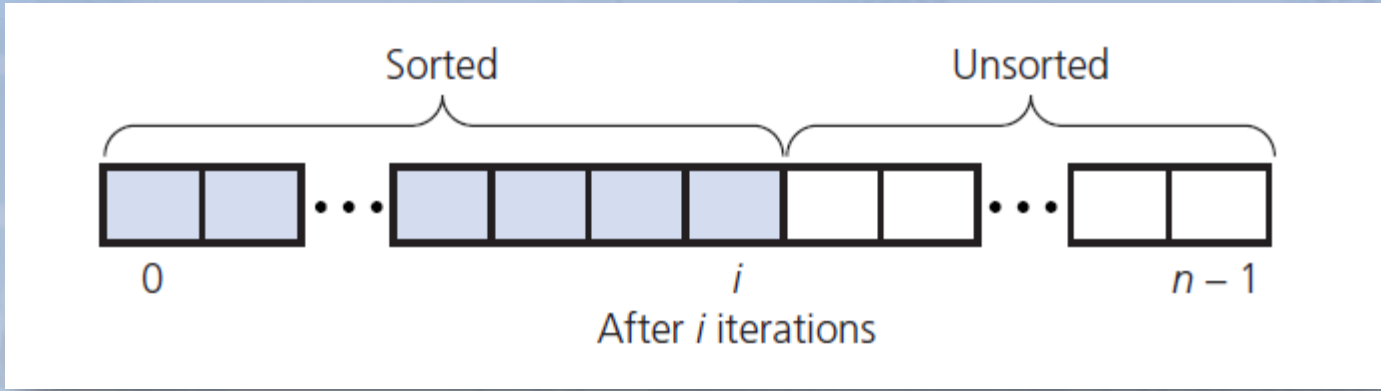

FIGURE 11-3 An insertion sort partitions the array into two regions

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The Insertion Sort

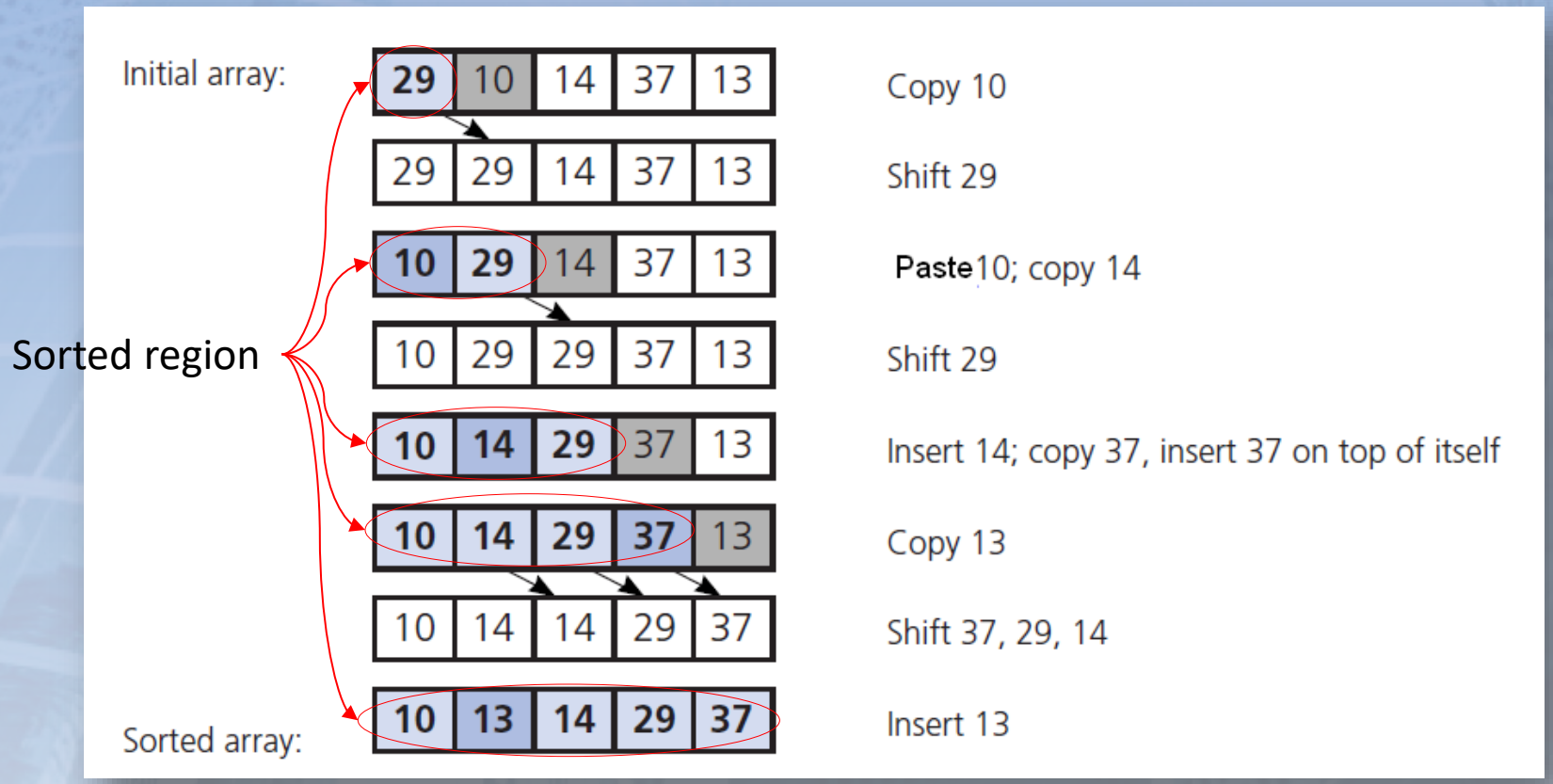

FIGURE 11-4 An insertion sort of an array of five integers

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The Insertion Sort

```
/** Sorts the items in an array into ascending order.
 @pre  None.
 @post  theArray is sorted into ascending order; n is unchanged.
 @param theArray  The given array.
 @param n  The size of theArray. */
template<class ItemType>
void insertionSort(ItemType theArray[], int n)
{
   // unsorted = first index of the unsorted region,
   // loc = index of insertion in the sorted region,
   // nextItem = next item in the unsorted region.
   // Initially,  sorted region is theArray[0],
   //             unsorted region is theArray[1..n-1].
   // In general, sorted region is theArray[0..unsorted-1],
   //             unsorted region theArray[unsorted..n-1]
```

LISTING 11-3 An implementation of the insertion sort

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The Insertion Sort

```
    for (int unsorted = 1; unsorted < n; unsorted++)
    {
        // At this point, theArray[0..unsorted-1] is sorted.
        // Find the right position (loc) in theArray[0..unsorted]
        // for theArray[unsorted], which is the first entry in the
        // unsorted region; shift, if necessary, to make room
        ItemType nextItem = theArray[unsorted];
        int loc = unsorted;
        while ((loc > 0) && (theArray[loc - 1] > nextItem))
        {
            // Shift theArray[loc - 1] to the right
            theArray[loc] = theArray[loc - 1];
            loc--;
        }  // end while
        // At this point, theArray[loc] is where nextItem belongs
        theArray[loc] = nextItem; // Insert nextItem into sorted region
    }  // end for
}  // end insertionSort
```

Note this is a nested loop for n

LISTING 11-3 An implementation of the insertion sort

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The Insertion Sort

- Analysis
  - Worst case $O(n^2)$
  - Best case (array already in order) is $O(n)$
- Unsuitable for large arrays
  - Unless already sorted
- Pros
  - Easy to understand
  - No additional space required
  - Preserves relative order of items with equal keys (stable)
- Cons
  - $O(n^2)$ – slow
  - Potentially suitable for low volume only

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Faster Sorting Algorithms

- The Merge Sort
- The Quick Sort
- The Radix Sort

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Merge Sort

- Divide and conquer sorting algorithm
- Halve the array, sort its halves, and then merge the halves

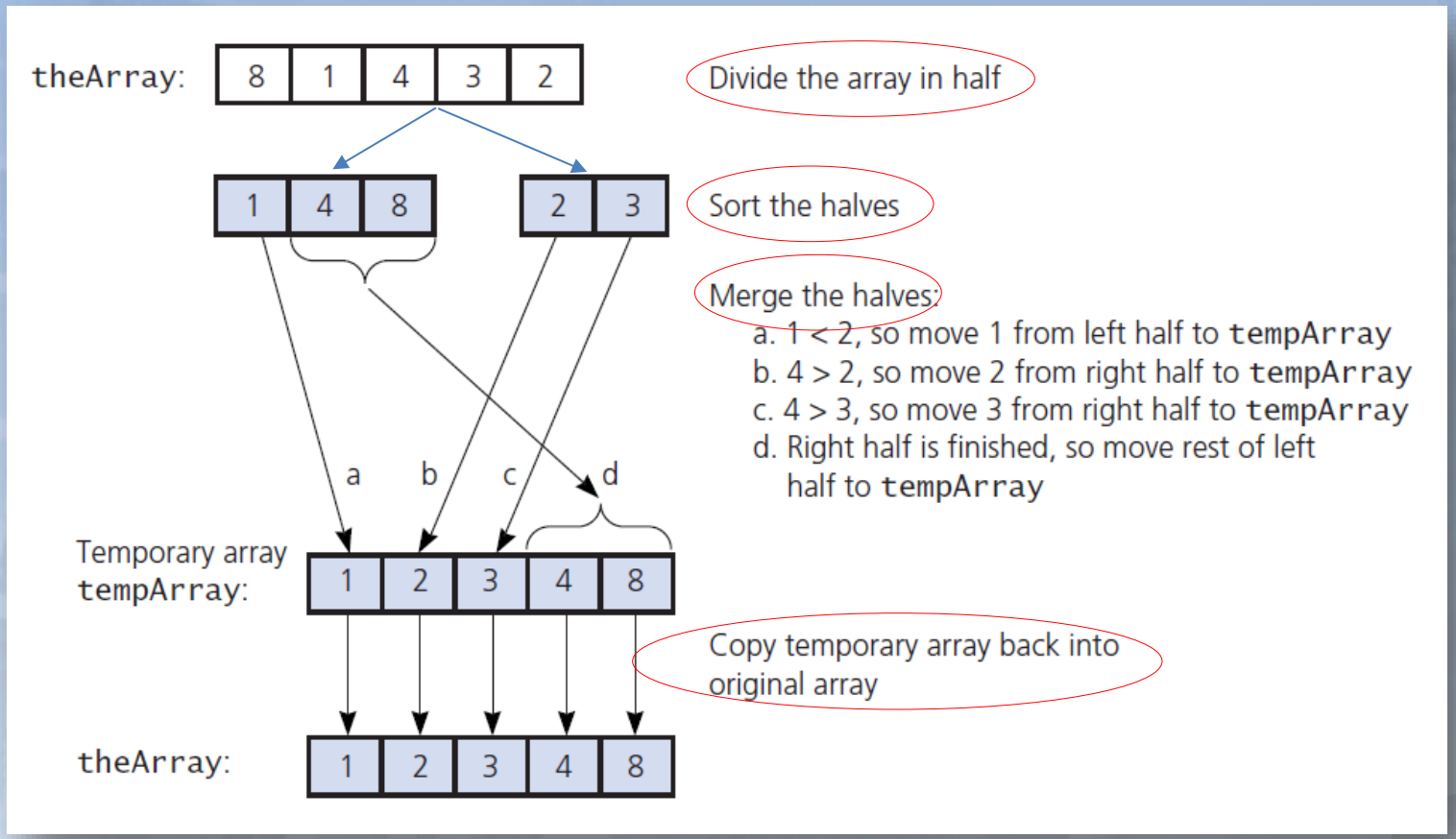

FIGURE 11-5 A merge sort with an auxiliary temporary array

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Merge Sort

```
// Sorts theArray[first..last] by
//    1. Sorting the first half of the array
//    2. Sorting the second half of the array
//    3. Merging the two sorted halves
mergeSort(theArray: ItemArray, first: integer, last: integer)
{
   if (first < last)
   {
      mid = (first + last) / 2        // Get midpoint

      // Sort theArray[first..mid]
      mergeSort(theArray, first, mid)

      // Sort theArray[mid+1..last]
      mergeSort(theArray, mid + 1, last)

      // Merge sorted halves theArray[first..mid] and theArray[mid+1..last]
      merge(theArray, first, mid, last)
   }
   // If first >= last, there is nothing to do
}
```

Pseudocode for the merge sort

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Merge Sort

- Analysis
  - O(n log n)
  - Worse case O(n log n)

- Pros
  - Performs well even on large number of items
  - Simple to implement
  - Preserves relative order of items with equal keys (stable)
- Cons
  - Requires additional memory O(n)
  - Requires copying of entries

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The Quick Sort

- Another divide-and-conquer algorithm
  - Choose a pivot
  - Partition an array into items that are
    - Less than or equal to the pivot and
    - Those that are greater than or equal to the pivot
  - Recursively apply the same process to the partitioned sub-arrays
  - Recursion stops when there is one element remaining in the sub-array

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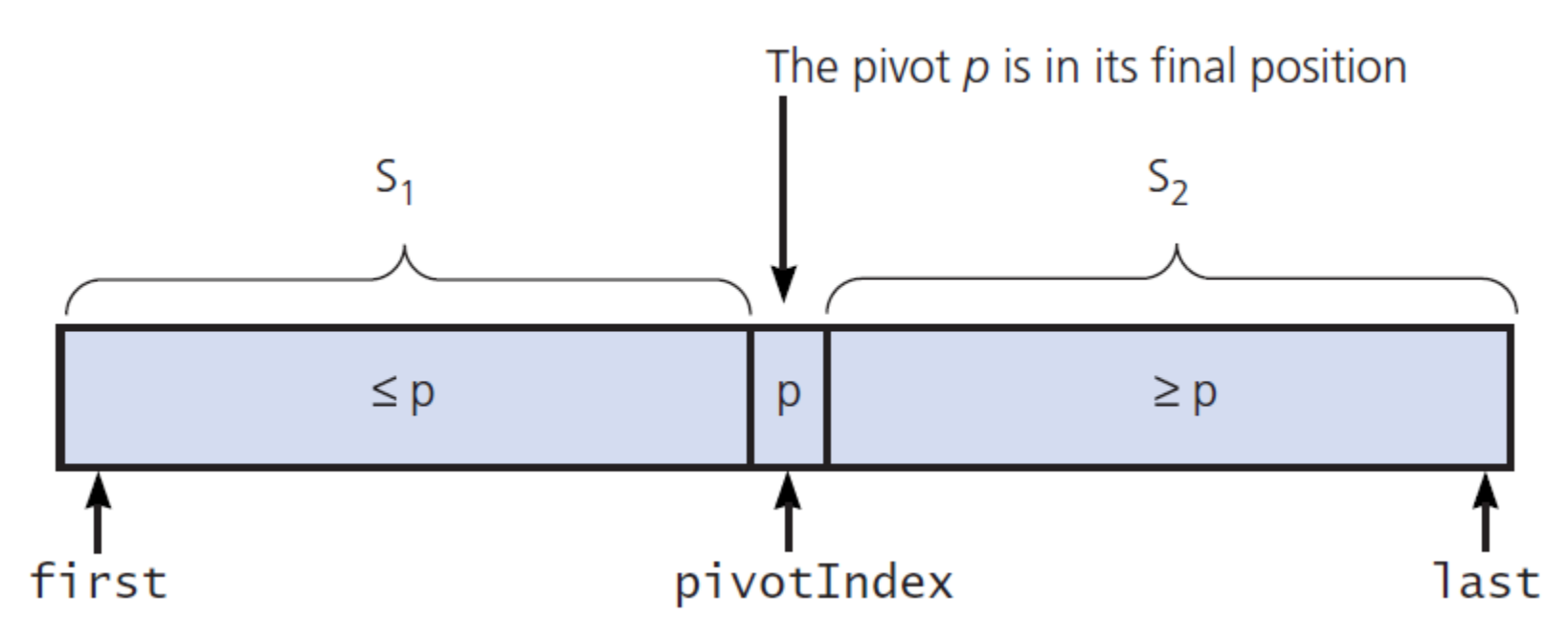
The Quick Sort



Quick sort algorithm

1. Choose a pivot
2. Partition the array
3. Recursively call

FIGURE 11-9 A partition about a pivot

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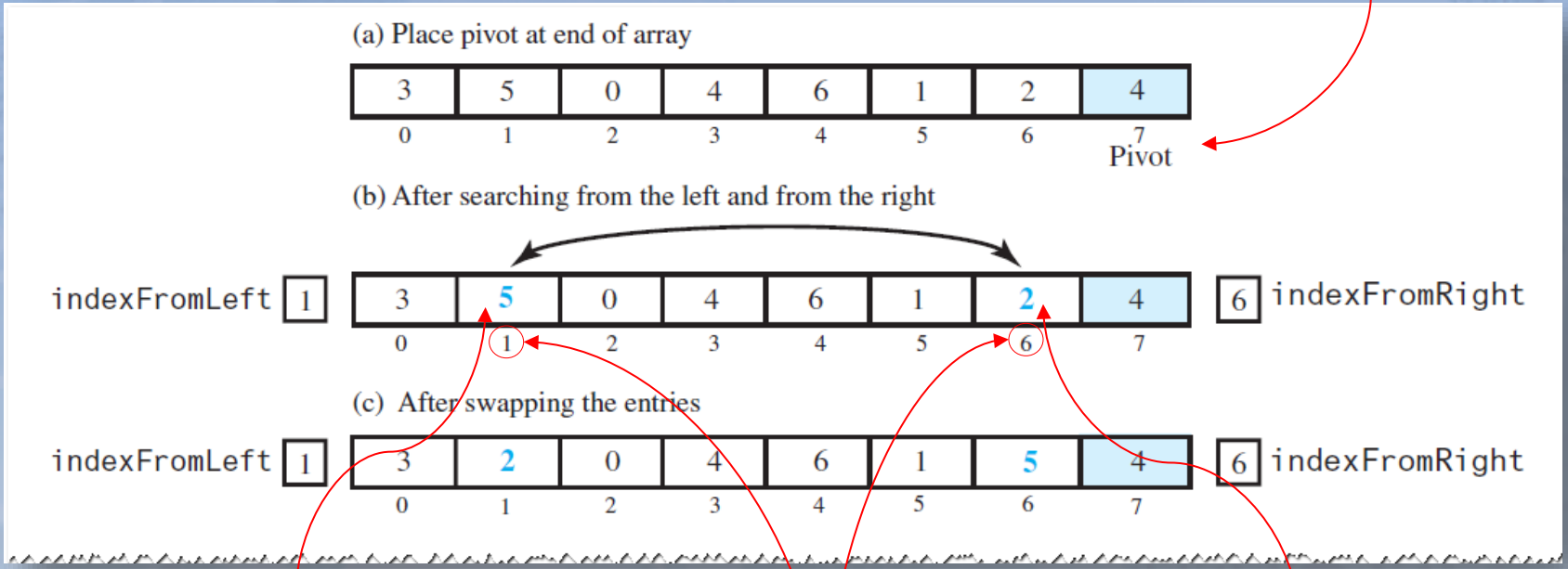
The Quick Sort

1. Choose a pivot



2. Start left to right, find 1st entry that is >= pivot

3. Start right to left, find 1st entry that is <= pivot

4. if indexFromLeft is less than indexFromRight, swap the two entries, else search ends

5. Continue left and right searches

FIGURE 11-10 A partitioning of an array during a quick sort

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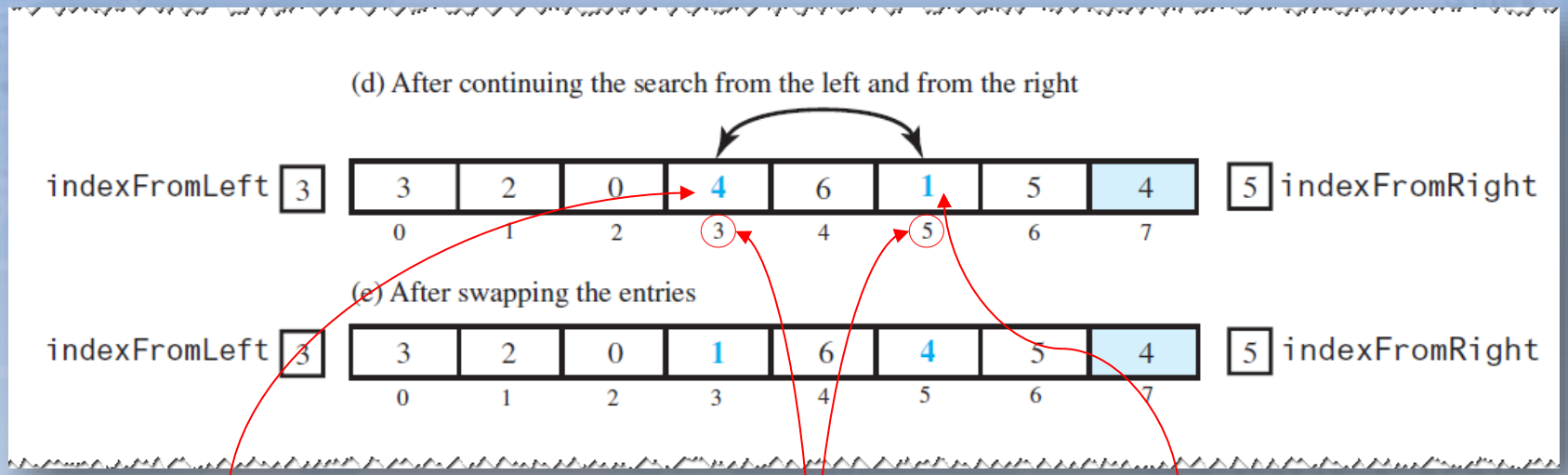
The Quick Sort



6. Find next entry that is >= pivot

7. Find next entry that is <= pivot

8. if indexFromLeft is less than indexFromRight, swap the two entries, else search ends

9. Continue left and right searches

FIGURE 11-10 A partitioning of an array during a quick sort

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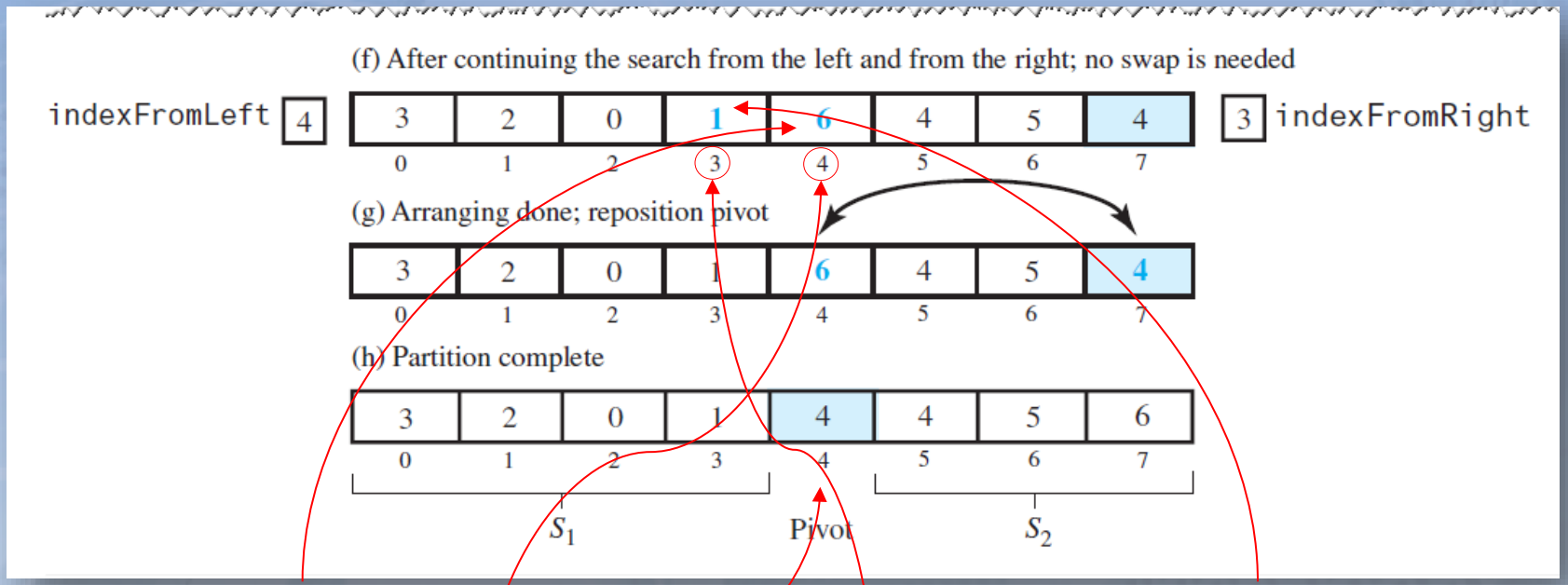
The Quick Sort



10. Find next entry that is >= pivot

11. Find next entry that is <= pivot

12. if indexFromLeft is less than indexFromRight, swap the two entries, else **search ends**

13. place the pivot between the subarrays S1 and S2 by swapping a[indexFromLeft] and a[last]; partition complete

14. Recursively call

FIGURE 11-10 A partitioning of an array during a quick sort

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The Quick Sort

- Pivot selection goal
  - Subarrays each have the same or nearly the same number of entries
- Pivot selection strategy
  - Last entry
  - First entry
  - Middle entry
  - Median-of-three

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The Quick Sort

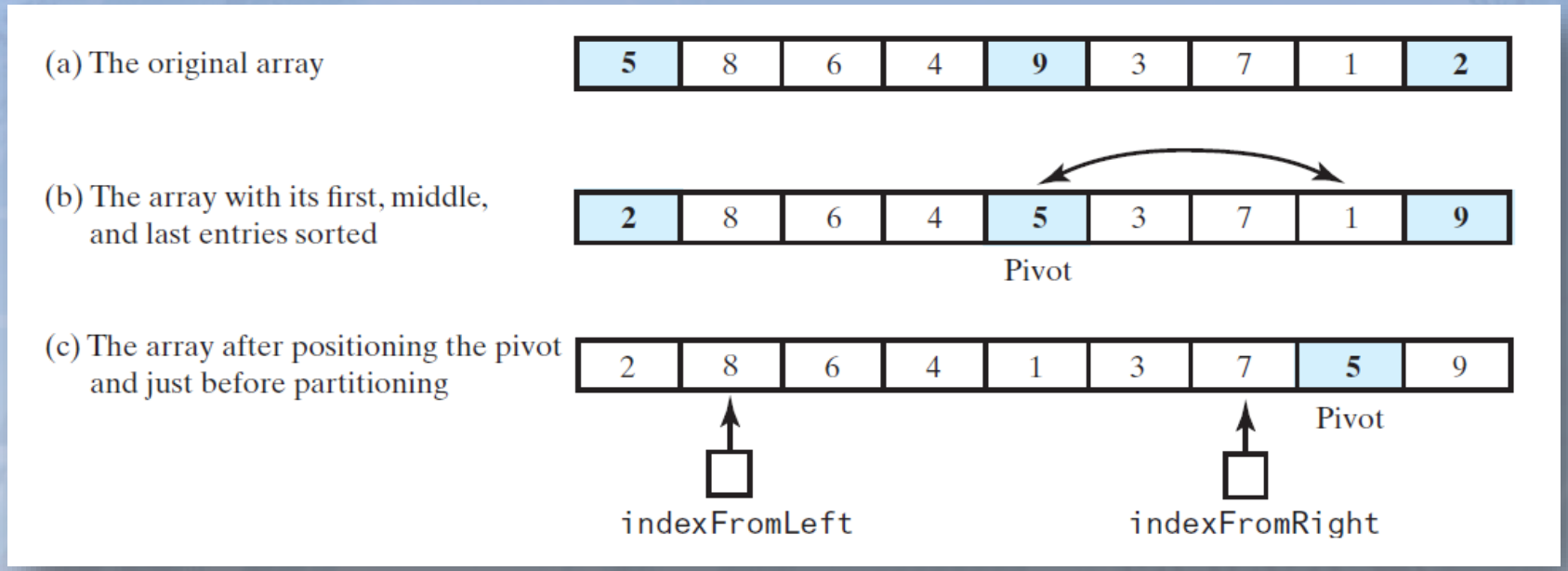

FIGURE 11-11 Median-of-three pivot selection

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The Quick Sort

- Analysis
  - Partitioning is an O(n) task
  - There are either $\log_2 n$ or 1 + $\log_2 n$ levels of recursive calls to quickSort.
- We conclude
  - Worst case O(n^2)
  - Average case O(n log n)
- Pros
  - Divide and conquer algorithm that is efficient on large number of items
  - No additional memory required
- Cons
  - Worst case O(n^2)
  - Does not preserve relative order of items with equal keys (not stable)

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The Radix Sort

- Different from other sorts
  - Does not compare entries in an array
- Begins by organizing data (say strings) according to least significant letters
  - Then combine the groups
- Next form groups using next least signficant letter

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The Radix Sort

0123, 2154, 0222, 0004, 0283, 1560, 1061, 2150	Original integers
(1560, 2150) (1061) (0222) (0123, 0283) (2154, 0004)	Grouped by fourth digit
1560, 2150, 1061, 0222, 0123, 0283, 2154, 0004	Combined
(0004) (0222, 0123) (2150, 2154) (1560, 1061) (0283)	Grouped by third digit
0004, 0222, 0123, 2150, 2154, 1560, 1061, 0283	Combined
(0004, 1061) (0123, 2150, 2154) (0222, 0283) (1560)	Grouped by second digit
0004, 1061, 0123, 2150, 2154, 0222, 0283, 1560	Combined
(0004, 0123, 0222, 0283) (1061, 1560) (2150, 2154)	Grouped by first digit
0004, 0123, 0222, 0283, 1061, 1560, 2150, 2154	Combined (sorted)

FIGURE 11-13 A radix sort of eight integers

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The Radix Sort

- Analysis
  - Requires *n* moves each time it forms groups
  - *n* moves to combine again into one group
  - Performs these $2 \times n$ moves *d* times
  - Thus requires $2 \times n \times d$ moves
  - More appropriate for chain of linked lists than for an array
- Radix sort is of order O(*n*)
- Pros
  - Radix sort is of order O(n)
  - Often faster than merge sort or quick sort for large number of items
  - Preserves relative order of items with equal keys (stable)
- Cons
  - Time complexity grows linearly with the number of digits
  - Requires same size keys, with standard way of breaking the keys into pieces

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A Comparison of Sorting Algorithms

	Worst case	Average case
Selection sort	n^2	n^2
Bubble sort	n^2	n^2
Insertion sort	n^2	n^2
Merge sort	$n \times \log n$	$n \times \log n$
Quick sort	n^2	$n \times \log n$
Radix sort	n	n
Tree sort	n^2	$n \times \log n$
Heap sort	$n \times \log n$	$n \times \log n$

FIGURE 11-14 Approximate growth rates of time required for eight sorting algorithms

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End

Chapter 11

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Appendix

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Merge Sort

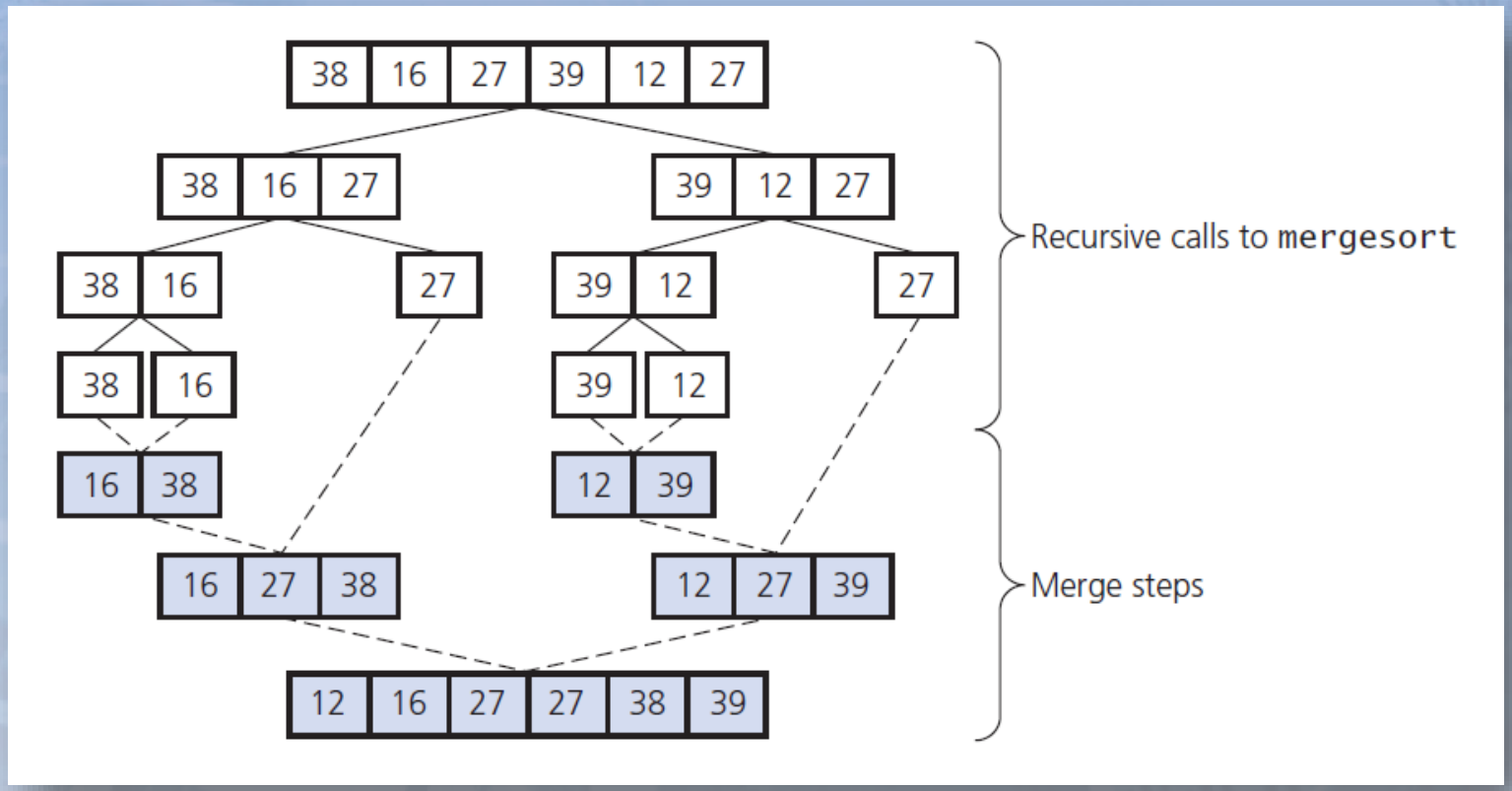

FIGURE 11-6 A merge sort of an array of six integers

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Merge Sort

```
const int MAX_SIZE = maximum-number-of-items-in-array;

/** Merges two sorted array segments theArray[first..mid] and
    theArray[mid+1..last] into one sorted array.
 @pre  first <= mid <= last. The subarrays theArray[first..mid] and
    theArray[mid+1..last] are each sorted in increasing order.
 @post  theArray[first..last] is sorted.
 @param theArray  The given array.
 @param first  The index of the beginning of the first segment in
    theArray.
 @param mid  The index of the end of the first segment in theArray;
    mid + 1 marks the beginning of the second segment.
 @param last  The index of the last element in the second segment in
    theArray.
```

LISTING 11-4 An implementation of merge and mergeSort

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Merge Sort

```
 @note  This function merges the two subarrays into a temporary
    array and copies the result into the original array theArray. */
template <class ItemType>
void merge(ItemType theArray[], int first, int mid, int last)
{
   ItemType tempArray[MAX_SIZE]; // Temporary array

   // Initialize the local indices to indicate the subarrays
   int first1 = first;           // Beginning of first subarray
   int last1 = mid;              // End of first subarray
   int first2 = mid + 1;         // Beginning of second subarray
   int last2 = last;             // End of second subarray

   // While both subarrays are not empty, copy the
   // smaller item into the temporary array
   int index = first1;           // Next available location in tempArray
   while ((first1 <= last1) && (first2 <= last2))
   {
```

LISTING 11-4 An implementation of merge and mergeSort

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Merge Sort

```
    while ((first1 <= last1) && (first2 <= last2))
    {
       // At this point, tempArray[first..index-1] is in order
       if (theArray[first1] <= theArray[first2])
       {
          tempArray[index] = theArray[first1];
          first1++;
       }
       else
       {
          tempArray[index] = theArray[first2];
          first2++;
       }    // end if
       index++;
    }  // end while
    // Finish off the first subarray, if necessary
    while (first1 <= last1)
```

LISTING 11-4 An implementation of merge and mergeSort

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Merge Sort

```
    while (first1 <= last1)
    {
       // At this point, tempArray[first..index-1] is in order
       tempArray[index] = theArray[first1];
       first1++;
       index++;
    }  // end while
    // Finish off the second subarray, if necessary
    while (first2 <= last2)
    {
       // At this point, tempArray[first..index-1] is in order
       tempArray[index] = theArray[first2];
       first2++;
       index++;
    }  // end for

    // Copy the result back into the original array
    for (index = first; index <= last; index++)
       theArray[index] = tempArray[index];
 }  // end merge
```

LISTING 11-4 An implementation of merge and mergeSort

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Merge Sort

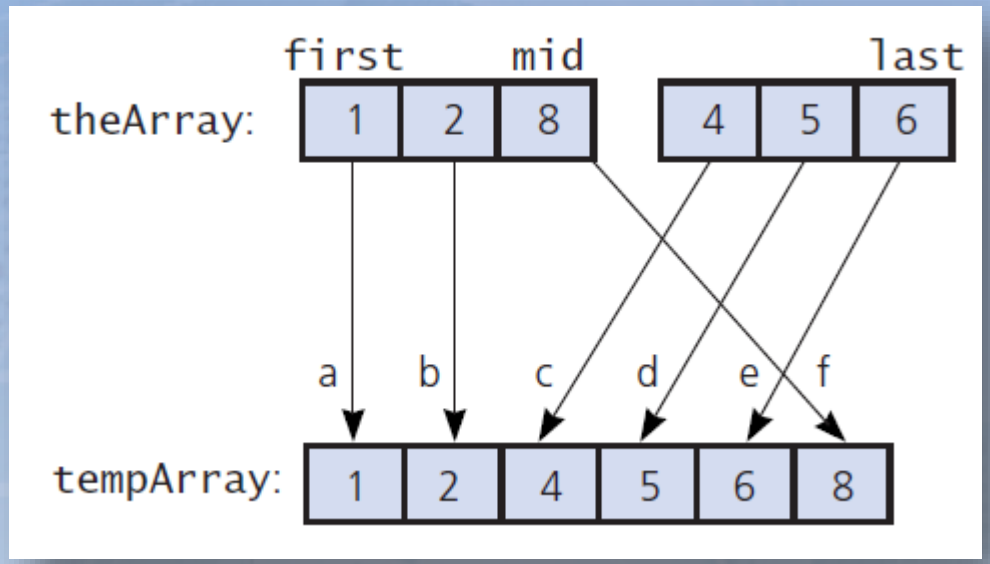

FIGURE 11-7 A worst-case instance of the merge step in a merge sort

Merge the halves:

a. 1 < 4, so move 1 from `theArray[first..mid]` to `tempArray`
b. 2 < 4, so move 2 from `theArray[first..mid]` to `tempArray`
c. 8 > 4, so move 4 from `theArray[mid+1..last]` to `tempArray`
d. 8 > 5, so move 5 from `theArray[mid+1..last]` to `tempArray`
e. 8 > 6, so move 6 from `theArray[mid+1..last]` to `tempArray`
f. `theArray[mid+1..last]` is finished, so move 8 to `tempArray`

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Merge Sort

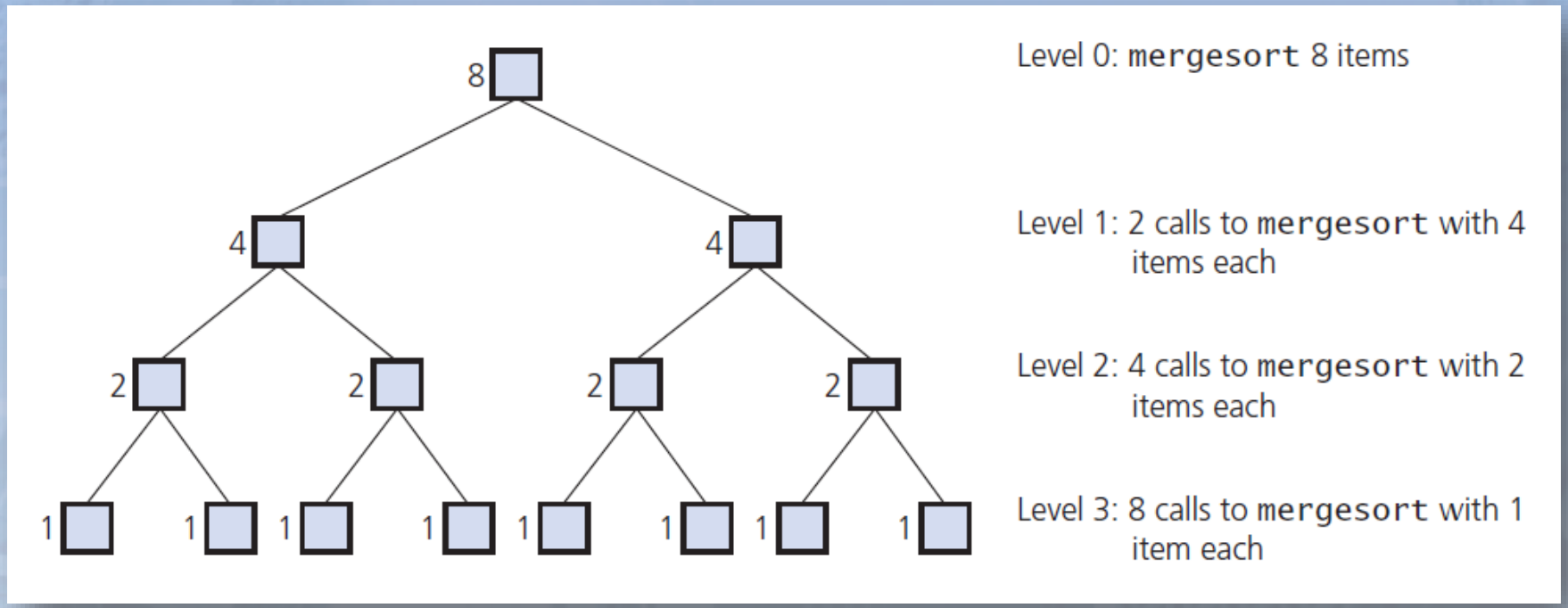

FIGURE 11-8 Levels of recursive calls to mergeSort, given an array of eight items

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The Quick Sort

```
// Sorts theArray[first..last].
quickSort(theArray: ItemArray, first: integer, last: integer): void
{
  if (first < last)
  {
    Choose a pivot item p from theArray[first..last]
    Partition the items of theArray[first..last] about p
    // The partition is theArray[first..pivotIndex..last]

    quickSort(theArray, first, pivotIndex - 1) // Sort S1

    quickSort(theArray, pivotIndex + 1, last)  // Sort S2
  }
  // If first >= last, there is nothing to do
}
```

First draft of pseudocode for the quick sort algorithm

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The Quick Sort

```
// Arranges the first, middle, and last entries in an array into ascending order.
sortFirstMiddleLast(theArray: ItemArray, first: integer, mid: integer,
                    last: integer): void
{
   if (theArray[first] > theArray[mid])
      Interchange theArray[first] and theArray[mid]

   if (theArray[mid] > theArray[last])
      Interchange theArray[mid] and theArray[last]

   if (theArray[first] > theArray[mid])
      Interchange theArray[first] and theArray[mid]
}
```

Adjusting the partition algorithm.

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The Quick Sort

```
// Partitions theArray[first..last].
partition(theArray: ItemArray, first: integer, last: integer): integer
{
   // Choose pivot and reposition it
   mid = first + (last - first) / 2
   sortFirstMiddleLast(theArray, first, mid, last)
   Interchange theArray[mid] and theArray[last - 1]
   pivotIndex = last - 1
   pivot = theArray[pivotIndex]

   // Determine the regions S1 and S2
   indexFromLeft = first + 1
   indexFromRight = last - 2

   done = false
   while (not done)
   {
      // Locate first entry on left that is ≥ pivot
```

Pseudocode describes the partitioning algorithm for an array of at least four entries

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The Quick Sort

```
done = false
while (not done)
{
   // Locate first entry on left that is ≥ pivot
   while (theArray[indexFromLeft] < pivot)
      indexFromLeft = indexFromLeft + 1

   // Locate first entry on right that is ≤ pivot
   while (theArray[indexFromRight] > pivot)
      indexFromRight = indexFromRight - 1

   if (indexFromLeft < indexFromRight)
   {
      Interchange theArray[indexFromLeft] and theArray[indexFromRight]
      indexFromLeft = indexFromLeft + 1
      indexFromRight = indexFromRight - 1
   }
   else
      done = true
}
// Place pivot in proper position between SecondS and Remark in array
```

Pseudocode describes the partitioning algorithm for an array of at least four entries

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The Quick Sort

```
            indexFromRight = indexFromRight - 1
        }
        else
            done = true
    }
    // Place pivot in proper position between S1 and S2, and mark its new location
    Interchange theArray[pivotIndex] and theArray[indexFromLeft]
    pivotIndex = indexFromLeft

    return pivotIndex
}
```

Pseudocode describes the partitioning algorithm for an array of at least four entries

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The Quick Sort

```
/** Sorts an array into ascending order. Uses the quick sort with
    median-of-three pivot selection for arrays of at least MIN_SIZE
    entries, and uses the insertion sort for other arrays.
 @pre  theArray[first..last] is an array.
 @post  theArray[first..last] is sorted.
 @param theArray  The given array.
 @param first  The index of the first element to consider in theArray.
 @param last  The index of the last element to consider in theArray. */
template <class ItemType>
void quickSort(ItemType theArray[], int first, int last)
{
   if ((last - first + 1) < MIN_SIZE)
   {
      insertionSort(theArray, first, last);
   }
```

LISTING 11-5 A function that performs a quick sort

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The Quick Sort

```
    }
    else
    {
        // Create the partition: S1 | Pivot | S2
        int pivotIndex = partition(theArray, first, last);

        // Sort subarrays S1 and S2
        quickSort(theArray, first, pivotIndex - 1);
        quickSort(theArray, pivotIndex + 1, last);
    }  // end if
}  // end quickSort
```

LISTING 11-5 A function that performs a quick sort

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The Quick Sort

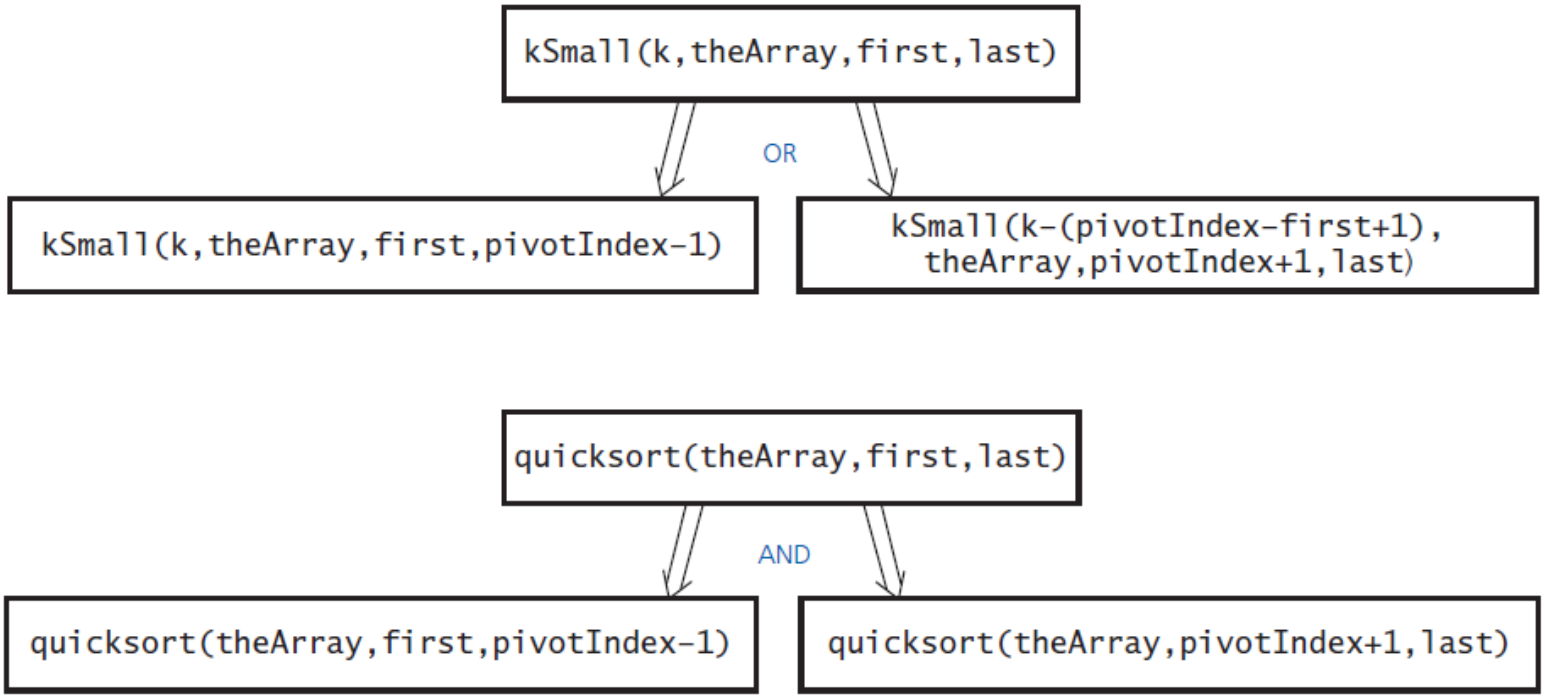

FIGURE 11-12 kSmall versus quickSort

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The Radix Sort

```
// Sorts n d-digit integers in the array theArray.
radixSort(theArray: ItemArray, n: integer, d: integer): void
{
   for (j = d down to 1)
   {
      Initialize 10 groups to empty
      Initialize a counter for each group to 0
      for (i = 0 through n - 1)
      {
         k = jth digit of theArray[i]
         Place theArray[i] at the end of group k
         Increase kth counter by 1
      }
      Replace the items in theArray with all the items in group 0,
        followed by all the items in group 1, and so on.
   }
}
```

Pseudocode for algorithm for a radix sort of *n* decimal integers of *d* digits each:

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